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An LLM-based agentic system for greenwashing detection☆

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ABSTRACT

Environmental reporting has assumed growing importance in capital markets, particularly in shaping investors' decision-making. However, the absence of universally accepted reporting standards and domain-specific assurance gives managers substantial discretion to present their sustainability performance in a deceptively favorable way, which is commonly called greenwashing. This risk is further compounded by investors' limited access to robust data and specialized analytical expertise, which impairs their ability to detect greenwashing in environmental disclosures. Large Language Models (LLMs) with advanced reasoning capabilities and integrated web search functions offer a novel approach to addressing these challenges. This study proposes a framework for developing LLM-based agentic systems to detect greenwashing risks in environmental disclosures. Since greenwashing is not a binary outcome but a multi-dimensional phenomenon, this study introduces seven indicators to assess different facets of greenwashing risks. These indicators are further integrated into a dashboard tailored to the needs of various users. This study contributes to the greenwashing literature by establishing a comprehensive framework for constructing LLM-based agents to detect greenwashing and proposing a set of quantifiable greenwashing indicators. It also contributes to the accounting profession by providing a tool for real-time monitoring of greenwashing risks.

1. Introduction

Environmental disclosures, such as Environmental, Social, and Governance (ESG) reports, are used to demonstrate companies' environmental performances and to communicate with stakeholders and the broader public. Within those reports, companies claim their sustainability positions, outline future ambitions and strategic plans, and report on their performance in the previous year. However, in most jurisdictions, environmental disclosures remain voluntary and are not held to the same legal standards as financial statements (Jiang et al., 2023). The absence of harmonized frameworks[1] permits companies to cherry-pick the metrics that present their environmental performance in the most favorable light (Gu et al., 2026). This practice is commonly referred to as

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[1] Commonly used ESG reporting frameworks include GRI, SASB, TCFD, CSRD, and IFRS-ISSB.

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"greenwashing." Furthermore, unlike financial statements, ESG reports are rarely subject to rigorous, independent audits. When assurance is provided, it is typically "limited," due to the inherent complexity of ESG disclosures and the limitations of current audit methodologies (Gu et al., 2023). Those challenges make greenwashed reports difficult to identify. From the investor's perspective, the volume, technical complexity, and inconsistent formatting of environmental data often exceed the capacity of investors to detect greenwashing risks effectively (Jiang et al., 2023). Without environmental expertise or a clear understanding of what data to look for, instances of greenwashing may remain hidden from investors.

Research has demonstrated the effectiveness of LLMs in financial statement analysis (Kim et al., 2024a), corporate risk assessment in financial reports (Kim et al., 2023), and helping investors to process information (Kim et al., 2024b). These capabilities highlight the potential of LLMs to process and interpret complex information in environmental disclosures. The reasoning ability of LLMs can enable the identification of counterfactual statements that may be overlooked by investors but are critical for informed decision-making. Furthermore, the LLMs' ability to retrieve online information and analyze text can enhance the efficiency of extracting relevant data from third-party sources to verify the environmental performance presented in ESG disclosures. Moreover, current audit practices often rely heavily on the knowledge, experience, and professional judgment of environmental experts to assess greenwashing risks. LLMs have the potential to assist or replace these experts by systematically identifying risk factors that may be overlooked due to cognitive limitations, informational gaps, or time constraints.

This paper explores the potential of using LLMs' capacities for textual analysis and online information retrieval to enhance greenwashing detection. Recent advancements in LLMs have opened new avenues for developing agentic systems capable of performing complex tasks in the financial and sustainability sectors (An et al., 2024; Schreyer et al., 2024; Vaghefi et al., 2025; Zou et al., 2025). This study investigates the feasibility of using LLM-based agentic systems to assess greenwashing risks through the analysis of publicly available corporate disclosures and online resources.

We adopt the Design Science Research Methodology (DSRM) (Hevner et al., 2004; Peffers et al., 2007), based on Geerts' (2011) six-phase application of DSRM in accounting information systems: (1) identifying and motivating the problem, (2) defining the objectives of a solution, (3) designing and developing the artifact, (4) demonstrating its use, (5) evaluating its performance, and (6) communicating the results.

The primary objective of this paper is to develop a comprehensive framework for designing and developing LLM-based agentic systems to conduct firm-specific greenwashing risk analysis based on predefined indicators and to generate corresponding risk dashboards. Drawing on the six categories of greenwashing risks proposed by Gu et al. (2026), this study introduces seven qualitative and quantitative indicators to guide the system design and implementation. For example, to detect the omission of unfavorable information, a common tactic in greenwashing, the agent can cross-reference ESG reports with external sources to uncover discrepancies or missing disclosures. These indicators not only facilitate greenwashing detection but also provide the foundation for constructing broader greenwashing indexes that can be employed in future empirical research. Those indicators are then processed by LLM-based agentic systems to extract relevant textual data from various sources and generate outputs that populate a greenwashing risk dashboard. To illustrate system application, a prototype system is developed using the *Shell* case. By reading *Shell*'s environmental-related public disclosures, including ESG and annual reports, and searching for relevant online information, the system generates responses to questions about greenwashing indicators. These responses are then manually verified and presented in a comprehensive dashboard.

This paper contributes to the accounting and greenwashing literature by proposing a novel framework for constructing LLM-based agentic systems and demonstrating their ability to detect greenwashing risks. Rather than measuring greenwashing from a single perspective, this study evaluates it across multiple dimensions, including environmental goals, initiatives, performance, and departmental activities, using state-of-the-art technologies. It further advances the literature by highlighting the critical role of large, exogenous data sources in enhancing the assurance of non-financial disclosures, such as ESG reports.

For accounting practice, the proposed LLM-based agentic systems offer a valuable tool for regulators to identify and investigate firms that may pose a higher risk of misleading environmental claims. By analyzing discrepancies between a company's disclosed environmental commitments and external data sources, regulators can detect potential greenwashing practices. This approach enables regulators to prioritize enforcement actions, conduct targeted audits, and ensure compliance with environmental reporting standards. Additionally, it can support the development of stricter regulations and improve transparency in corporate environmental disclosures, ultimately enhancing accountability and protecting investors from misleading ESG reporting. Furthermore, the greenwashing dashboard is designed to accommodate diverse user needs. For example, it can extract and display key statistics from environmental disclosures, reducing the need for manual review of source documents. It also can provide comparative analysis by allowing investors to select peer firms and display their corresponding indicators side by side to support decision-making and enhance environment-oriented portfolio strategies. By providing a data-driven analysis, the LLM-based agentic systems can equip auditors with valuable insights for risk assessment.

The next section of this paper reviews the literature on greenwashing and LLM applications in accounting and ESG and identifies the problems and objectives of this project (DSRM phases 1 and 2). Section 3 proposes a comprehensive framework for developing LLM-based agentic systems capable of detecting greenwashing risks (DSRM phase 3). Section 4 discusses the construction of greenwashing indicators, and Section 5 illustrates the framework through a case study using a real-world example of alleged greenwashing (DSRM phases 4 and 5). Section 6 discusses the limitations of this study and suggests ideas for future research. The final section offers conclusions. Overall, we communicate the findings of our research on developing an LLM-based agentic system for greenwashing detection through this paper (DSRM phase 6).

2. Background and literature review

2.1. Greenwashing

Greenwashing remains a widespread and growing concern. Over the past six years, greenwashing incidents have been most prevalent in Europe and North America. According to RepRisk (2024), of the 3,868 companies associated with misleading communication, 2,213 (57 percent) were also linked to environmental issues that constitute greenwashing. When regulators uncover greenwashing practices, they often impose penalties due to the significant harm to investors and the environment. For example, in April 2024, the SEC secured a judgment against Volkswagen Group of America Finance, LLC for misleading investors about emissions compliance in its $8 billion bond offerings, resulting in over $48 million in financial penalties.[2] In November 2024, the SEC fined Invesco Advisers, Inc. $17.5 million for misrepresenting ESG integration and including non-ESG assets in its reported figures without a clear policy.[3]

Undetected greenwashing can significantly hinder climate progress and distort capital allocation. It not only undermines the integrity of environmental disclosures but also distorts the signaling function of environmental reporting, misleading investors, consumers, and regulators, and generating inefficiencies in financial and environmental decision-making.[4] However, the detection and mitigation of potential fraudulent elements in environmental disclosures, specifically greenwashing, remains underexplored. Among the limited studies, Zhang (2023) finds that enhanced governmental environmental monitoring significantly mitigates greenwashing practices. Zhang (2024) explores the role of Artificial Intelligence (AI) in effectively curbing corporate greenwashing. Dao et al. (2024) note that existing LLM-based greenwashing detection methods, primarily based on BERT models (Kim et al., 2023), are still in their early stages. Gleiser et al. (2024) propose the use of LLMs to identify discrepancies in sentiment between sustainability reports and actual financial media coverage. Their approach offers a novel perspective by integrating sustainability reports with online information. Leveraging the capabilities of LLMs, our study aims to explore their application in detecting greenwashing risks and generating risk reports.

Gu et al. (2026) classify six forms of greenwashing in environmental disclosures: 1) misconduct, 2) selective disclosure, 3) misclassification, 4) hollow promise, 5) in name only, and 6) misleading presentation. Misconduct refers to situations in which a firm alters data or portrays environmental efforts in an inaccurate way. Selective disclosure involves omitting material environmental information that would weaken a firm's sustainability profile. Misclassification means portraying harmful activities as environmentally sound or blending them with genuine sustainability initiatives. Hollow promise refers to making environmental commitments without taking meaningful steps to carry them out. "In name only" captures cases in which a firm establishes sustainability units or committees that serve no substantive role. Misleading presentation captures the use of stylistic or framing techniques in environmental disclosures that guide stakeholders toward a more favorable view of a firm's environmental record. This classification lays the foundation for developing greenwashing indicators within the system proposed in this paper.

2.2. LLM and agentic systems

An LLM is a type of natural language processing (NLP) model based on deep learning, primarily used for text generation, text comprehension, automated conversation, and machine translation. Its core technologies include neural networks, probabilistic statistics, and large-scale data training, which collectively enable the simulation of human language comprehension and expression. Vaswani et al. (2017) introduce a transformer architecture, which serves as the backbone for modern LLMs. Brown et al. (2020) introduce the GPT-3 model and the concept of few-shot learning, in which users provide a small number of task examples in the input prompt for the model to learn, significantly enhancing AI's capabilities in natural language processing and drawing widespread attention to LLMs. Over time, LLMs have evolved to include models such as GPT-4o by OpenAI, DeepSeek-R1 by High-Flyer, Gemini 1.5 by Google DeepMind, Claude 3 by Anthropic, and Llama 3 by Meta, each demonstrating significant performance improvements across different tasks. These models represent advancements in multimodal capabilities, reasoning, cost efficiency, and safety, contributing to a more diverse landscape of AI-powered applications. LLMs offer advanced capabilities in text analysis, online information retrieval, and reasoning, allowing them to identify potential risks that might be overlooked by humans. In addition, LLMs' contextual understanding capability can effectively capture semantic relationships (Siano, 2025). These models can process vast amounts of unstructured data, extract meaningful patterns, and highlight potential anomalies in financial reports, thereby contributing to enhanced audit quality and efficiency.

One of the primary applications of LLMs in auditing is their ability to summarize financial reports efficiently (Dong et al., 2024; Yang et al., 2024; Kim et al., 2024c). By distilling large volumes of text into concise and meaningful summaries, LLMs enable auditors to grasp key insights quickly from financial statements, management discussions, and footnotes. Levich and Knust (2025) show that LLMs can reliably condense and extract key information from lengthy corporate disclosures, demonstrating their value for handling unstructured financial text. Huang et al. (2023) develop FinBERT, a state-of-the-art LLM that adapts to the finance domain, and show that it incorporates finance knowledge and can summarize contextual information in financial texts better than other LLMs. Kim et al., (2024b) show that when a document has a positive (negative) sentiment, its summary by LLMs becomes more positive (negative),

[2] https://www.sec.gov/enforcement-litigation/litigation-releases/lr-25969.
[3] https://www.sec.gov/newsroom/press-releases/2024–179.
[4] https://environment.ec.europa.eu/topics/circular-economy/green-claims_en.

indicating that LLMs add considerable value for investors with information-processing constraints. Kim et al. (2023) find that LLMs are effective in corporate risk assessment. This capability is particularly useful in risk-based audits, where auditors must focus on high-risk areas without being overwhelmed by excessive documentation. Compared to financial analysts, LLMs outperform in their ability to predict directional earnings changes (Kim et al., 2024a). Bai et al. (2023) highlight LLMs as valuable tools for uncovering hidden insights in corporate disclosures. These studies suggest that LLMs can detect deceptive language, identify financial distress indicators, and flag unusual reporting behaviors that may indicate fraud or misstatements.

Another critical function of LLMs is their potential role in audit planning and purpose-driven analysis. Fotoh and Mugwira (2025) find that LLMs can handle routine tasks effectively, including producing human-like text and drafting audit documentation and reports. Serving as co-pilots for auditors in audit tasks (Gu et al. 2024c), LLMs can assist in developing risk assessments, planning audit procedures, and identifying key areas that require substantive testing. These models can prioritize areas for audit focus based on financial trends, regulatory changes, and historical anomalies, allowing auditors to allocate resources more effectively (Gu et al., 2024a). Moreover, LLMs excel in reasoning and complex analysis, making them suitable for evaluating the implications of financial data and regulatory disclosures. Unlike traditional rule-based audit systems, LLMs can interpret nuanced relationships between textual and numerical data, drawing inferences. For instance, the capacity of LLMs to integrate external data sources, such as macroeconomic indicators, industry benchmarks, and regulatory filings, enhances auditors' ability to contextualize financial information within broader economic trends (Gu et al., 2024a).

Current advancements in LLMs have opened new possibilities for agentic systems that leverage the LLMs to automate complex tasks in financial and ESG domains (An et al., 2024; Schreyer et al., 2024; Stratopoulos and Wang, 2025). An et al. (2024) introduce FinVerse, a meticulously crafted agentic system designed to address a broad range of financial topics. By integrating over 600 financial application programming interfaces (APIs), the system enables access to more accurate and extensive financial information. Schreyer et al. (2024) propose an agentic auditing framework that demonstrates how an LLM agent can autonomously perform audit procedures by planning and adapting its actions based on intermediate outcomes, illustrating the feasibility of agent-led workflows in assurance contexts.

In practice, audit firms have recently started to adopt agentic AI solutions. For example, KPMG uses AI agents in the Clara platform to automate routine tasks, support decision-making, and help auditors focus on high-risk areas.[5] Deloitte uses Zora AI, an agentic AI that perceive and execute accounting functions.[6] EY leverages the EY.ai Agentic Platform that can respond to real-time events, adapt to regulatory changes, and help decision-making.[7] PwC's agent OS is an enterprise AI command center that connects and scales intelligent agents into business.[8] Technology firms and regulatory bodies are also actively engaging with agentic AI advancements to establish a more automatic and intelligent working environment for the accounting profession (AuditBoard, 2024; Diligent, 2024; Ho, 2023).

2.3. LLM applications in environmental disclosures

A growing body of research explores the use of LLMs to analyze the environmental section in ESG reports (Dong et al., 2024). Extracting actionable insights from ESG reports remains a challenge due to their lack of standardization, the prevalence of dense technical content, and considerable length, which hinder effective interpretation and analysis (Gu et al., 2024b). Gu et al. (2024b) demonstrate the difficulty of information extraction by comparing the performance of ChatGPT to simple natural language processing (i.e., keyword search). They find that while ChatGPT demonstrates higher specificity due to its ability to evaluate text contextually, it demonstrates lower sensitivity, making it more prone to false negatives. Bao and Gullkvist (2024) find that ChatGPT can extract information on greenhouse gas (GHG) emissions to help external auditors assure ESG reports. These capabilities have the potential to address the challenge of greenwashing detection in which companies often make broad or unverifiable sustainability claims in public communications.

Recent studies also explore the applications of LLM-based agents in the ESG domain. Zou et al. (2025) apply a retrieval-augmented LLM agent, ESGReveal, to extract ESG metrics from narrative corporate reports, significantly outperforming traditional NLP methods in structured data extraction. Vaghefi et al. (2025) develop a climate finance agent to track investments in climate adaptation by applying multi-step reasoning. These approaches provide a technical and conceptual foundation for developing LLM agents that can autonomously search, reason, and cross-validate environmental disclosures, enabling scalable detection of greenwashing fraud.

Overall, the growing emphasis on ESG reporting has intensified concerns about greenwashing, where firms make exaggerated or misleading claims about their sustainability practices. A key challenge lies in the absence of comprehensive, real-time monitoring tools capable of identifying greenwashing risks, which hinders investors' ability to distinguish genuine ESG initiatives from misleading disclosures. Addressing this gap motivates our research and highlights the risks posed by greenwashing not only to investors and regulators but also to the credibility of auditing and assurance processes (DSRM phase 1). The primary objective of this research is to develop a comprehensive framework for constructing agentic systems that can systematically conduct firm-specific greenwashing risk

[5] https://kpmg.com/us/en/media/news/kpmg-clara-smart-audit-platform.html.

[6] https://www2.deloitte.com/us/en/pages/about-deloitte/articles/press-releases/deloitte-unveils-zora-ai-agentic-ai-for-tomorrows-workforce.html.

[7] https://www.ey.com/en_gl/newsroom/2025/03/ey-launching-ey-ai-agentic-platform-created-with-nvidia-ai-to-drive-multi-sector-transformation-starting-with-tax-risk-and-finance-domains.

[8] https://www.pwc.com/us/en/about-us/newsroom/press-releases/pwc-launches-ai-agent-operating-system-enterprises.html.

analysis using LLMs (DSRM phase 2). The framework will incorporate predefined indicators of greenwashing, operationalize them into measurable variables, and organize the results into a set of risk dashboards.

3. A framework for developing LLM-based agentic systems to detect greenwashing risks

This section presents the design of a comprehensive framework for developing LLM-based agentic systems to detect greenwashing risks (DSRM phase 3). The framework encompasses five key dimensions: 1) data sources, 2) LLM capabilities, 3) prompt engineering, 4) greenwashing indicators, and 5) results presentation (shown in Fig. 1). It leverages LLM capabilities to cross-reference diverse data sources, apply structured reasoning, and deliver findings in formats tailored to users' needs.

The data dimension in Fig. 1 includes various data sources to be ingested by LLM-based agentic systems to enable a comprehensive assessment of greenwashing risks. These sources include ESG reports, financial reports, earnings call transcripts, greenwashing cases investigated by the SEC, sensor data on temperature and emissions from the Environmental Protection Agency (EPA) and other environmental agencies, and other online information. By combining both structured and unstructured data from multiple sources, the systems can capture inconsistencies between a company's public statements and its actual sustainability performance. The diversity of data sources enhances the system's ability to detect misleading claims that may not be evident in a single document.

The LLM capability dimension highlights the essential functions that the agentic systems must perform to analyze greenwashing risks effectively, including dynamic knowledge, reasoning, and online search. Dynamic knowledge allows the systems to adapt to evolving ESG regulations and industry trends. Reasoning helps the systems to critically evaluate claims, rather than simply summarizing reports. Online search enables real-time retrieval of external information to verify corporate statements. Without these capabilities, the systems would be limited to historical disclosures, reducing their ability to detect emerging greenwashing tactics.

A crucial aspect of optimizing the performance of LLM-based agentic systems is prompt engineering. The framework emphasizes the need for convergence of results, ensuring that similar queries produce stable and accurate conclusions. Temperature[9] settings control is an important factor in controlling the variability of responses, balancing factual accuracy with flexibility in interpreting environmental disclosures. Setting the temperature to 0 enhances the reliability and consistency of LLMs when analyzing complex, often ambiguous sustainability claims.

The indicator dimension encompasses both qualitative and quantitative measures for assessing greenwashing. Drawing on the six types of greenwashing risk practices outlined by Gu et al. (2026), specific indicators are developed for each category, taking industry-specific and firm-specific characteristics into account (see Section 4 for details on indicator development). Qualitative indicators, such as expert evaluations and linguistic heuristics, help to identify exaggerated or misleading ESG claims. Conversely, quantitative indicators rely on numbers and statistical benchmarks, such as detecting discrepancies between reported carbon emissions and independent third-party estimates. The integrated qualitative and quantitative indicators could enhance the systems' ability to detect greenwashing practices with greater precision and contextual relevance.

Finally, the presentation dimension defines how agentic systems communicate their findings. Given the diverse needs of stakeholders, such as regulators, investors, and auditors, the systems should support multiple output formats. For example, text summaries can be used to convey concise qualitative insights suitable for regulatory review, while visualizations can offer investors an intuitive illustration of greenwashing risk levels and enable firm-to-firm comparisons. The choice of presentation format is critical since different stakeholder groups will interpret and act on ESG insights in distinct ways. Tailoring the outputs to the targeted audience would enhance the usefulness and clarity of the systems.

Fig. 2 demonstrates the main components of the LLM-based agentic system based on the aforementioned framework. This system composes three phases, including preparation, development, and use. The preparation phase contains indicator development and disclosure collection. Experts develop greenwashing risk indicators that capture various forms of greenwashing and can be evaluated by LLMs. This phase also contains the collection of relevant corporate disclosures and supporting data from financial and non-financial sources for each indicator. For example, ESG reports and annual reports are necessary to cross-verify ESG claims and identify instances of selective disclosure, while emissions data should be retrieved from authoritative sources such as the EPA to validate reported figures. Users can either upload ESG reports themselves or select a firm for the system to retrieve the company's documents automatically from disclosure databases.

In the development phase, an LLM agent is then deployed in the system to analyze environmental disclosures and generate insights aligned with the predefined greenwashing risk indicators. For each indicator, experts construct a dedicated prompt that includes a detailed description, processing steps, and an output format. The description specifies the required data sources, defines the expected output, and stipulates the model's temperature setting, typically set to zero to ensure consistent, reproducible results.[10] Through its integrated online search and advanced reasoning capabilities, the LLM agent can analyze greenwashing risks by cross-referencing ESG claims against external information sources, which improves the accuracy and depth of its evaluations. The agent produces an initial result for each indicator that includes a greenwashing risk assessment, supported by relevant evidence, and classifies the risk into one of four levels: high, moderate, low, or uncertain (if the LLM is not able to find relevant information), and also provides a confidence

[9] In the context of LLM agentic systems, temperature is a hyperparameter that controls the randomness, creativity, and determinism of the agent's responses and decisions. The lower the temperature, the more deterministic, focused, and consistent the agent will be. Default temperature is usually 1.0, providing a balance of average creativity and predictability.

[10] Since the online-information and LLMs will update, the results generated might not be the same over time. Therefore, the results will be updated periodically.

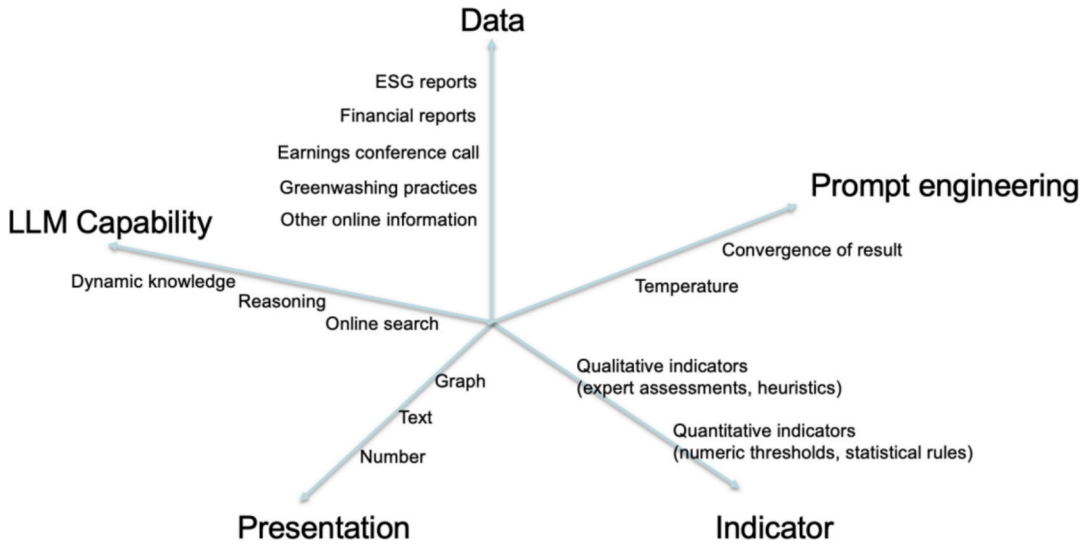


Fig. 1. The framework for developing LLM-based agentic systems to detect greenwashing risks.

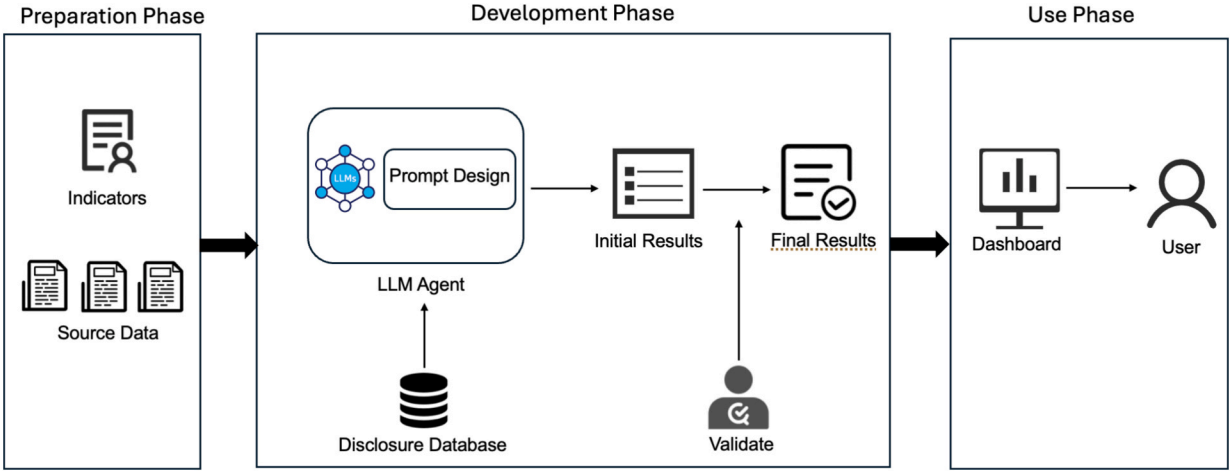


Fig. 2. An LLM-based agentic system for greenwashing detection.

level reflecting how strongly the evidence supports the assessment. Next, human verification is required to validate the initial results of the LLM agent. This process involves reviewing the source references and evaluating the logical consistency of the LLM's reasoning. It also helps to identify which data sources are most informative for specific indicators. If the results are too general, the prompts may be iteratively refined to improve their reliability.

Finally, in the use phase, a dashboard can be generated using the verified results, and the presentation of these outputs can be adjusted to match users' formatting requirements. This dashboard can be deployed as a web-based interface connected to a backend database, offering a comprehensive view of greenwashing indicators for individual firms. Stakeholders and researchers can leverage these insights for further analysis, such as exploring ESG disclosures, comparing firms to industry peers, and assessing compliance with sustainability standards. The streamlined interface allows for easy interaction with system outputs, facilitating the interpretation of ESG performance across organizations. This approach enhances transparency and supports data-driven decision-making in greenwashing risk analysis. Users will continuously monitor system behavior and conduct regular evaluations of the greenwashing detection results. Any identified biases or unfair outcomes will be documented and reported, enabling ongoing refinement and improvement of the system.

4. Greenwashing indicators

Drawing on the six categories of greenwashing risks identified by Gu et al. (2026), we develop a set of qualitative and quantitative indicators to be used in our greenwashing detection system, along with their corresponding data inputs, as outlined in Table 1. These indicators are designed to capture various forms of greenwashing.

To identify the misconduct risk, our LLM-based agentic system can retrieve relevant data from authoritative third-party sources and use it to verify the information disclosed in ESG reports. For example, the system can examine greenhouse gas (GHG) emissions by comparing data disclosed in ESG reports with those recorded by the EPA to detect discrepancies in reported figures.

For the selective disclosure risk, our LLM-based agentic system can detect this risk by flagging material contingencies (e.g., adverse environmental events) that appear in regulatory filings (8-K/6-K) or are reported by reputable external sources (e.g., Reuters) but are absent from concurrent ESG disclosures or annual reports. This requires cross-referencing ESG reports, financial reports, 8-K/6-K filings, and external news or data.

The misclassification risk occurs when a company inaccurately categorizes ESG-related financial information, such as classifying a non-environmental investment as an environmental initiative, to enhance the perceived quality of its sustainability performance (Gu et al., 2026).[11] This risk can be detected by our LLM-based agentic system via cross-verification between ESG reports and financial statements to ensure consistency between reported ESG metrics and underlying financial data. This process helps to identify instances where sustainability-related expenditures and revenues are misclassified to inflate ESG performance. To verify misclassification, the systems must collect data from ESG reports, financial performance documents, Reuters, the EPA, and the SEC.

The hollow promises risk include two indicators: goal inconsistency and slow progress. Goal consistency can be identified by an LLM-based agentic system when a company downgrades its previously stated environmental goals, such as reductions in target amounts, extensions of timelines, or narrowing the scope of commitments. To assess slow progress, our LLM-based agentic system can analyze a company's historical performance to evaluate whether meaningful advancements have been made toward stated environmental goals by calculating the fulfillment percentage. These checks prevent firms from continuously postponing sustainability targets without meaningful progress.

For the "in name only" risk, our LLM-based agentic system can assess engagement levels of new sustainability departments by examining indicators such as the frequency and substance of updates on official company websites or the presence of public announcements. These checks help to determine whether the department is functionally contributing to sustainability efforts or serving a symbolic role. ESG reports, official company websites, and reputable news sources can help identify this risk.

To detect the misleading presentation risk, our LLM-based agentic system can analyze language patterns, categorizing content into hard information (substantial facts), gray information (aspirational statements), and other non-substantive content. This classification helps to assess the extent to which a company relies on aspirational language rather than presenting concrete evidence of ESG performance.

Based on the six types of indicators, our LLM-based agentic system can offer a systematic approach to detecting greenwashing by cross-verifying companies' ESG claims against independent sources. These indicators can enhance transparency and accountability by leveraging data from a mix of regulatory filings, financial reports, third-party databases, and online sources.

5. Implementation of an LLM-based agentic system for greenwashing detection

This section demonstrates the application of the proposed system to a real-world case and evaluates its performance (DSRM phases 4 and 5). Based on the framework and indicators discussed in Sections 3 and 4, we use the energy sector as an example to develop an LLM-based agentic system for greenwashing detection, as the sector is well-known for greenwashing in its environmental disclosures

[11] In this study, we define "ESG-related financial information" as any financial disclosure that links an ESG activity to a specific past, present, or future monetary value. We define an "environmental investment" as a capital allocation where the primary stated purpose is to improve environmental performance. Investments are classified as "non-environmental" if their primary purpose is related to core business expansion, marketing, financial engineering, or other activities not directly tied to environmental goals.

Table 1
Indicators for Greenwashing Risks.

Greenwashing risks	Indicators	Required data
Misconduct	Discrepancies between reported environmental data and information from authoritative third-party sources (e.g., GHG emissions disclosed in ESG reports diverge from data recorded by the EPA).	ESG reports, EPA emission data, GHG data
Selective disclosure	Undisclosed environmental risks (e.g., unfavorable events reported in 8-K filings or other documents are omitted from ESG reports)	ESG reports, financial performance, 8-K (6-K) filings, Reuters, EPA news, SEC press releases
Misclassification	Misclassification of ESG-related financial Information	ESG reports, financial statements, Reuters, EPA, SEC press releases
Hollow promise	Downgrading of previously stated environmental goals, including reductions in targeted amounts, extensions of timelines, or narrowing of scope	Past five-year ESG reports
	Minimal or significantly delayed progress toward stated environmental goals	ESG reports
In name only	Limited operational activity of sustainability departments (e.g., infrequent website updates, sparse news releases, or lack of publicly documented events)	ESG reports, firm's official website, reputable news source
Misleading presentation	Using vague, irrelevant, or immaterial content to create a misleading impression of environmental responsibility	ESG reports

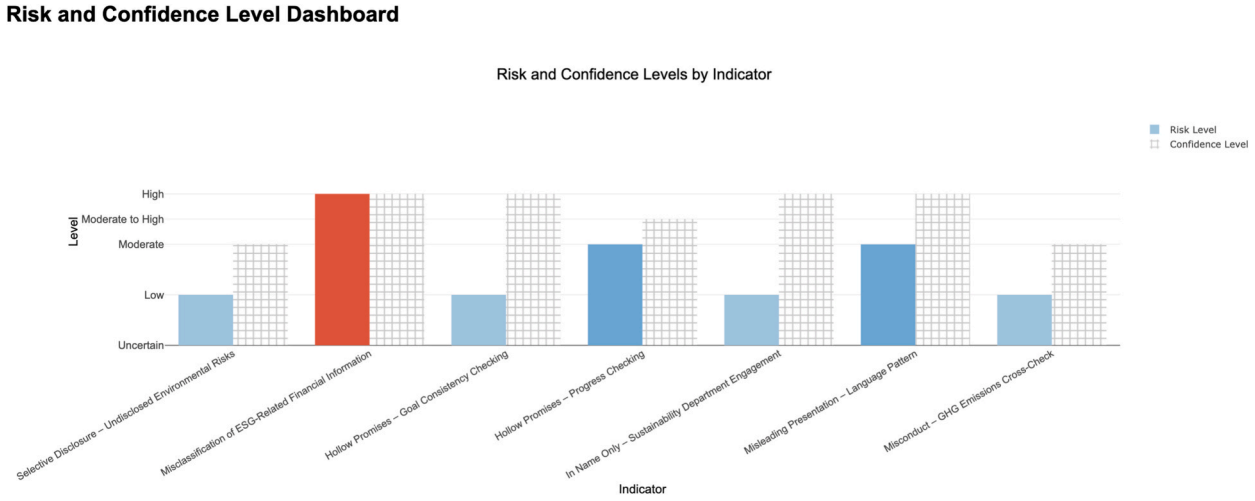


Fig. 3. Risk and confidence level by indicator.

(RepRisk, 2024). To demonstrate the application of the LLM-based agentic system, we use *Shell*, a major energy company, as a case study. Since 2023, Shell has faced public allegations of greenwashing its environmental reporting in its 2021 disclosures.[12] Our analysis is based on *Shell*'s 2021 sustainability report and annual report. Trusted public sources, such as Reuters, the EPA, and the SEC, are also incorporated to support cross-verification with *Shell*'s reports. The system consists of two main components: the LLM prompt design and the dashboard interface. These components are discussed in the following sections.

5.1. Prompt design

Two approaches can be used to prepare the LLMs for the agentic system: fine-tuning an existing model or using prompt engineering. Compared to fine-tuning a model, prompt engineering has advantages because it does not need model re-training, which would require expensive computing power and labeled datasets. Instead, it only involves crafting instructional queries to optimize the model's output. This approach enables dynamic integration of up-to-date information from online sources (e.g., financial data, news, and the websites of regulators and the EPA) without modifying the model. Prompt engineering provides fairly accurate results for various tasks (Vatsal & Dubey, 2024). Given its efficiency and cost-effectiveness, we adopt prompt engineering to configure the LLM component of the system. Specifically, we use OpenAI's most advanced ChatGPT model available at the time of writing, GPT-4o, to ensure high-quality performance in greenwashing detection.

The prompt is designed to include the system's designated role, a detailed description of greenwashing indicators, and the relevant

[12] https://globalwitness.org/en/campaigns/fossil-fuels/shell-faces-groundbreaking-complaint-for-misleading-us-authorities-and-investors-on-its-energy-transition-efforts/?gad_source=1&gad_campaignid=12467226173&gbraid=0AAAADm6LOBMucsh8AUN7kgRisSHWnOQX&gclid=Cj0KCQjwucDBBhDxARIsANqFdr24lQycM7tDZjoeuFh0D7athESIvImEpEnQ-nN1QETyeZdIeOyRm58aApjxEALw_wcB.

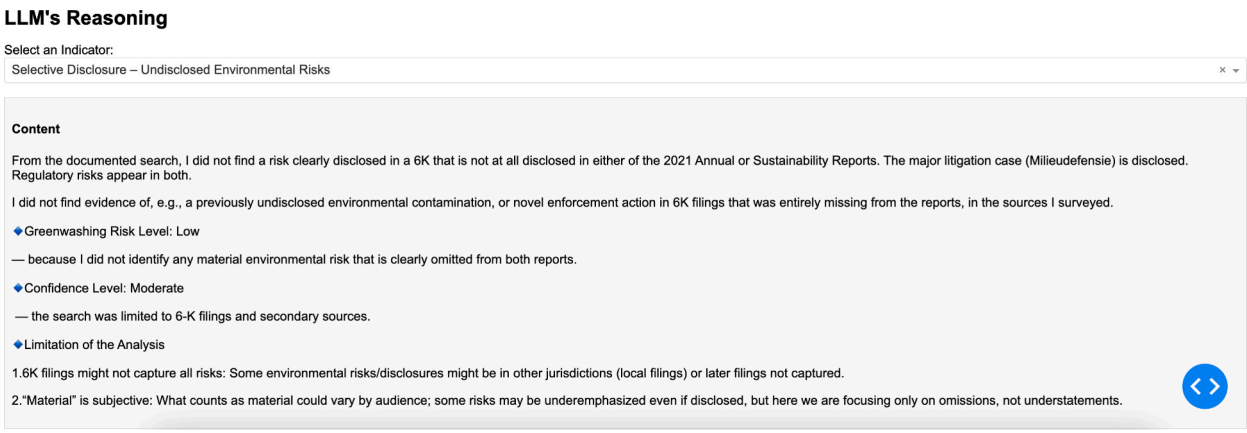


Fig. 4. Undisclosed environmental risks.

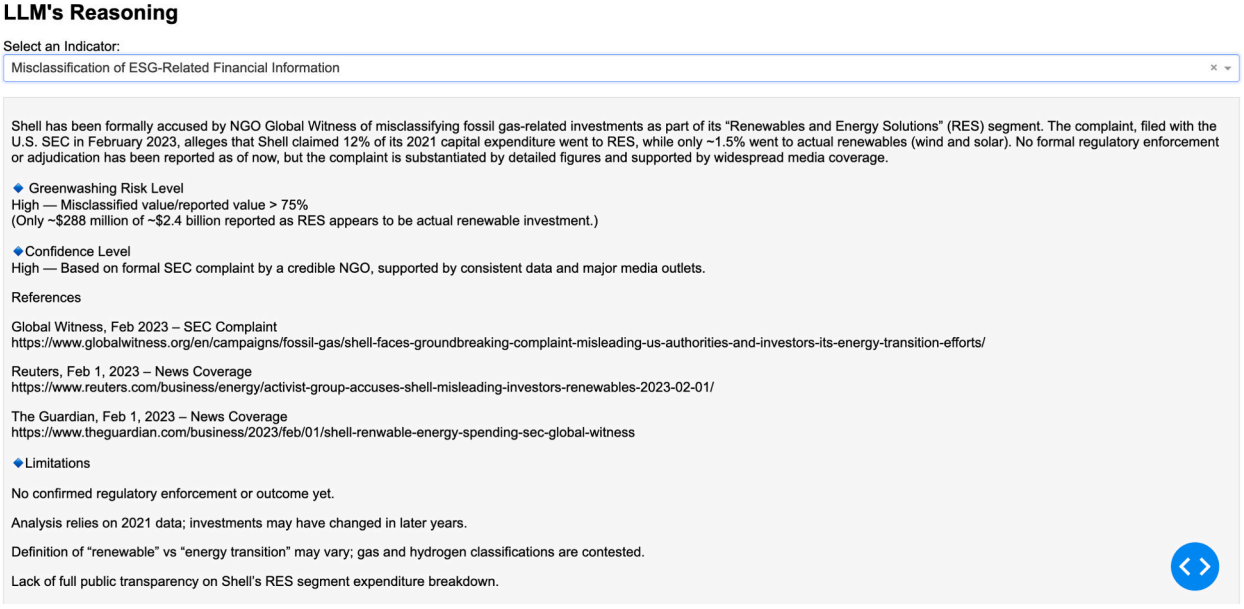


Fig. 5. Misclassification of ESG-related financial information.

data sources. Prompts classify greenwashing risk into four levels[13]: Low, Moderate, High, or Uncertain. A Low rating signifies minimal to no risk, a Moderate rating indicates a medium risk level, and a High rating denotes a significant, evidence-supported risk requiring close scrutiny. An Uncertain rating is assigned when insufficient data prevents assessment. Each indicator is evaluated against its own customized risk threshold. Each assessment requires the LLM to provide a rationale for the assigned risk level, along with a "confidence level"[14] to reflect the reliability of the output. Detailed specifications for the prompts, along with the corresponding risk and confidence level definitions for each indicator, are provided in the Appendix.

5.2. Dashboard

After executing the designed prompts, the LLM-based agentic system exports the results for each greenwashing indicator into Word and Excel files. We validate those results and then populate them into a dashboard using Python. The initial section of the dashboard is

[13] The classification of risk levels as well as confidence levels presented later in the paper is user-defined and context-dependent. System developers can customize the thresholds for each level according to their specific needs and professional judgment.

[14] The initial version of the confidence level was generated using LLMs. Four researchers then reviewed and discussed this version, refining it based on their expert judgment. The finalized version was subsequently incorporated into the prompt.

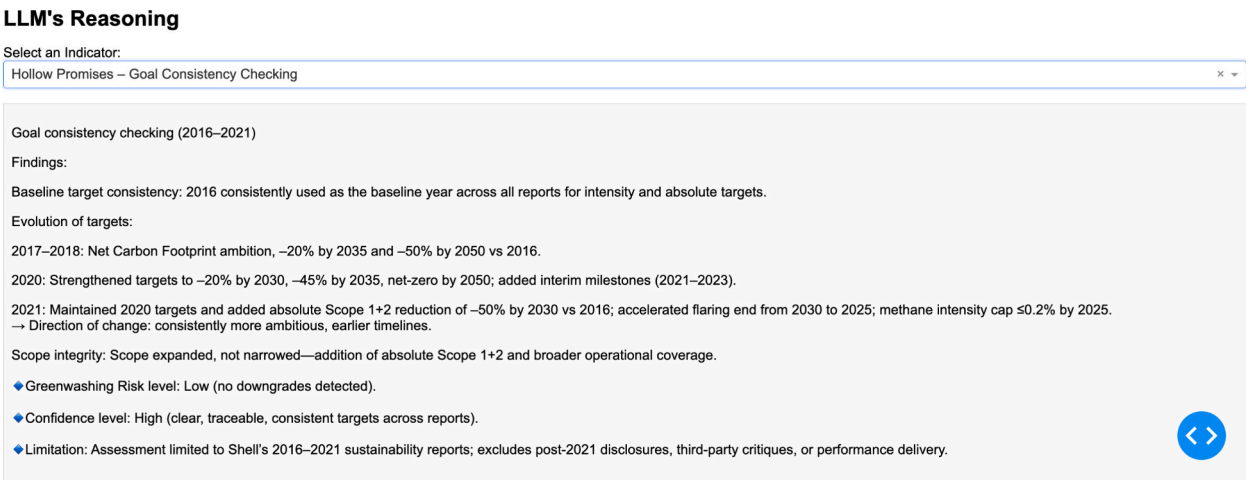


Fig. 6. Goal consistency checking.

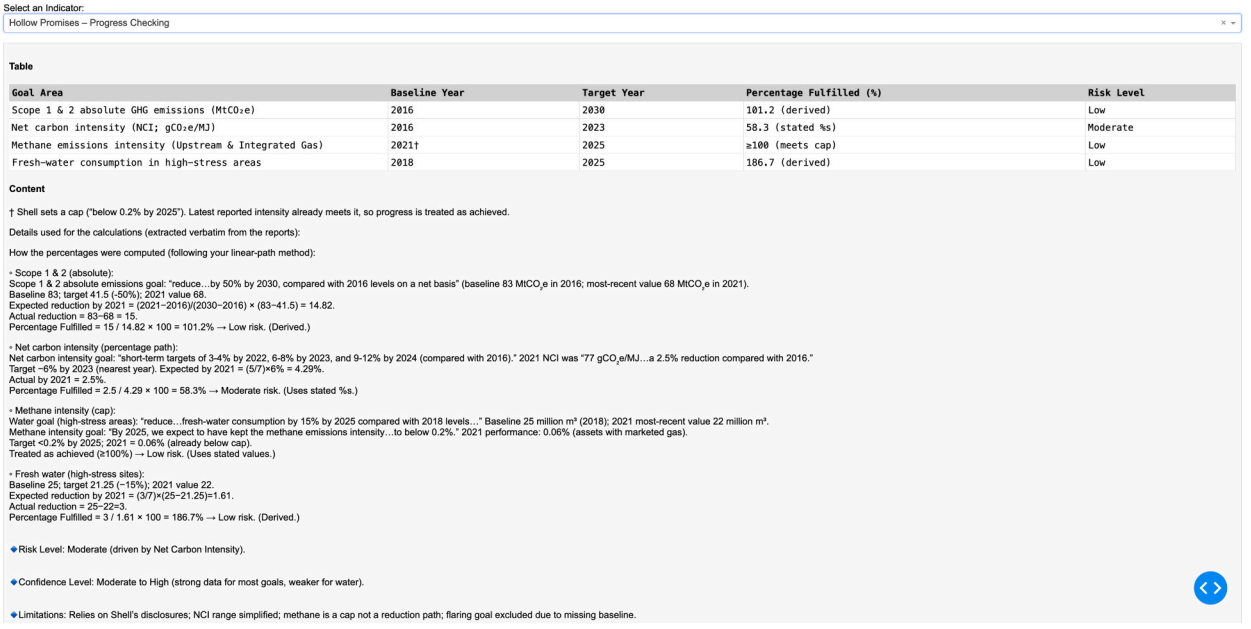


Fig. 7. Progress checking.

presented in Fig. 3. The bar chart displays the risk and confidence levels based on information gathered from relevant sources. Among the seven indicators, one is categorized as high risk (highlighted in red), two as moderate risk, and four as low risk. Each indicator is also assigned a confidence level that reflects the system's degree of certainty in its risk assessment.

The next section of the dashboard presents the LLM's reasoning behind the greenwashing risk assessment. It includes a dropdown list that displays the supporting evidence for each indicator. For example, Fig. 4 shows the analysis results for undisclosed environment risks. Since *Shell* is a foreign company, the LLM-based agentic system searched its Form 6-K filings to identify environmental risks and compare them to the information in its annual report and sustainability report. The results show that GPT did not find a risk clearly disclosed in a 6-K that was not disclosed in either of the 2021 Annual or Sustainability Reports. The major litigation case (Milieu-defensie) is disclosed. Therefore, according to the indicator's definition, this does not constitute an omission. As a result, the risk of greenwashing is considered low. The confidence level is moderate because the review was limited to available 6-K filings and secondary sources. It is important to note that the analysis is limited to disclosures in annual reports, sustainability reports, and Form 6-K filings, and may not capture environmental risks that are unreported or underreported.

Fig. 5 presents the findings regarding the potential misclassification of ESG-related financial information. The system identifies one instance of alleged misclassification, as reported by Global Witness (2023), which GPT evaluates as potentially high risk. Multiple

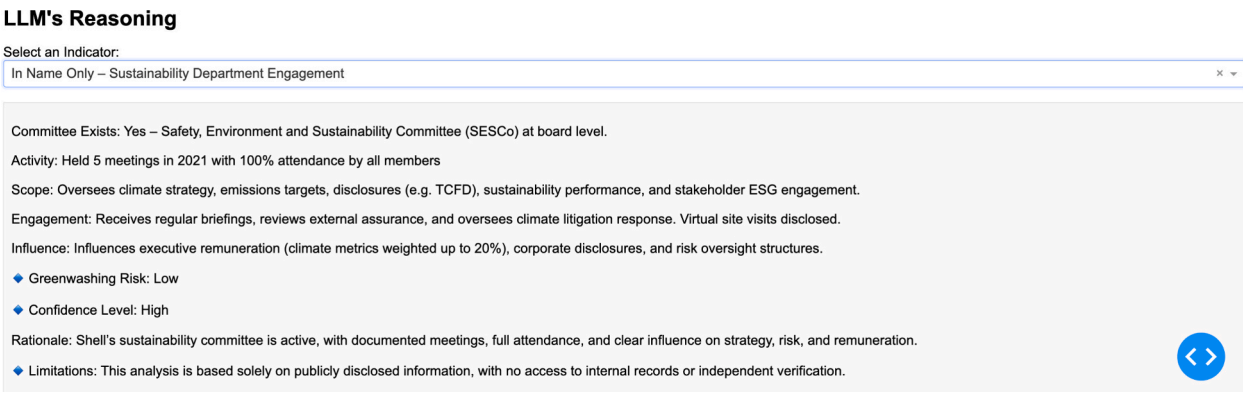


Fig. 8. Sustainability department engagement.

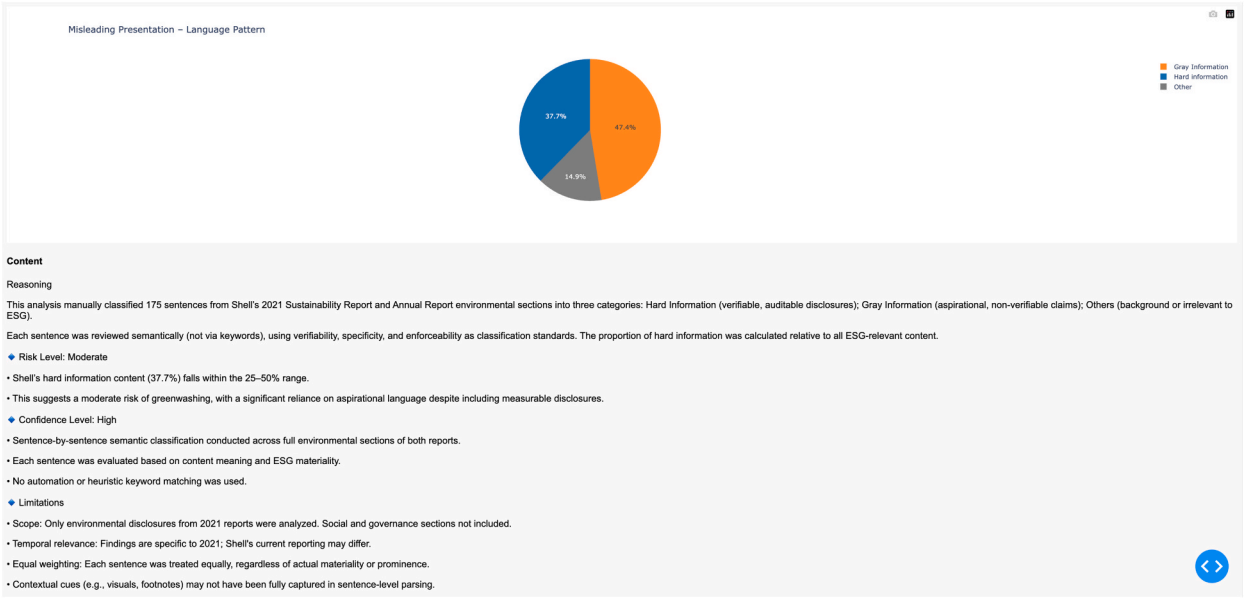


Fig. 9. Language pattern.

external sources support this assessment, indicating a high level of confidence in the evaluation. However, GPT also acknowledges the limitations of its analysis, including the absence of confirmation from the SEC and its lack of access to internal data on actual capital allocation. This analysis is based on publicly available information and does not assert or imply any legal or regulatory violation. Allegations referenced here are attributed to their respective sources and remain under review or legal contest.

Fig. 6 and 7 present GPT-based analyses of *Shell*'s climate goal setting and progress. Fig. 6 presents the goal-consistency analysis of *Shell*'s sustainability reports from 2016 to 2021. It finds no evidence of downgrading; instead, the company has increased the ambition of its near- and medium-term targets. On this basis, the greenwashing risk is assessed as low with high confidence. Fig. 7 shows the results by extracting each environmental goal, standardizing its baseline and nearest target year, and calculating the Percentage of Goal Fulfillment. If a goal contains multiple-year targets, the nearest year target is used to compare the percentage fulfilled. Based on the fulfillment percentage, goals are risk-rated, and the company's overall disclosure is classified at the level of the highest-risk goal identified. The system assigns a moderate risk level to this indicator because the reduction target in Net Carbon Intensity has been achieved at only 58.3 %. Both analyses depend entirely on public disclosures, and no independent verification or access to internal performance data is conducted.

Fig. 8 presents the findings on Shell's sustainability department's engagement level. GPT identifies that *Shell* has a committee titled Safety, Environment and Sustainability (SESCo), which undertook activities such as climate-focused meetings with full attendance. This committee also reviews the strategy and oversees emission targets. Overall, the greenwashing risk associated with this indicator is assessed as low with a high level of confidence. However, this analysis is based solely on publicly disclosed information with no access to internal records or independent verification.

Fig. 9 presents a pie chart generated based on a language pattern analysis conducted by the LLM agentic system. It first identifies the

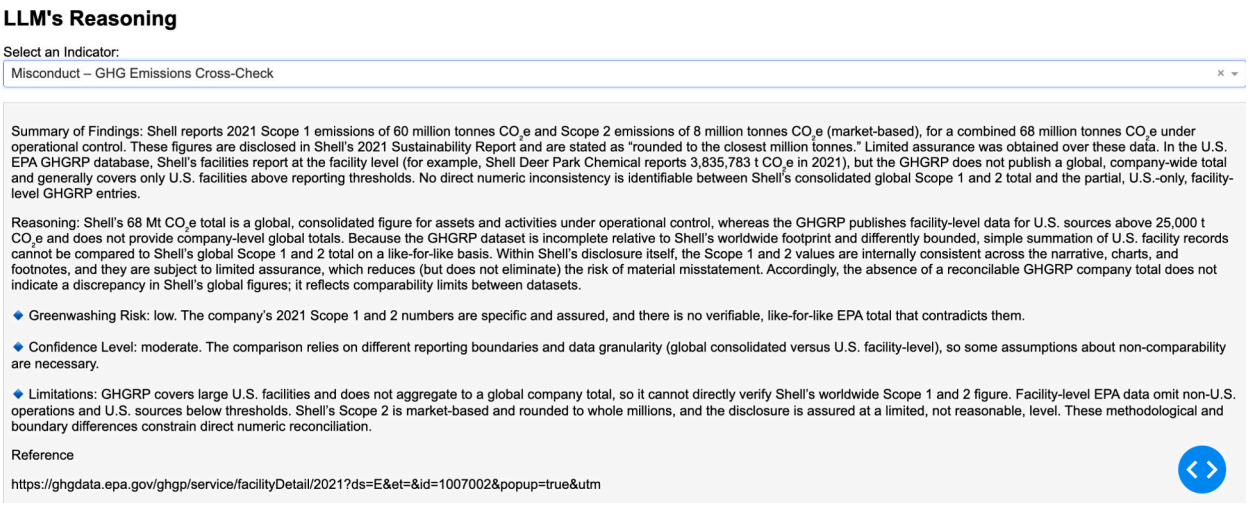


Fig. 10. GHG emissions cross-check.

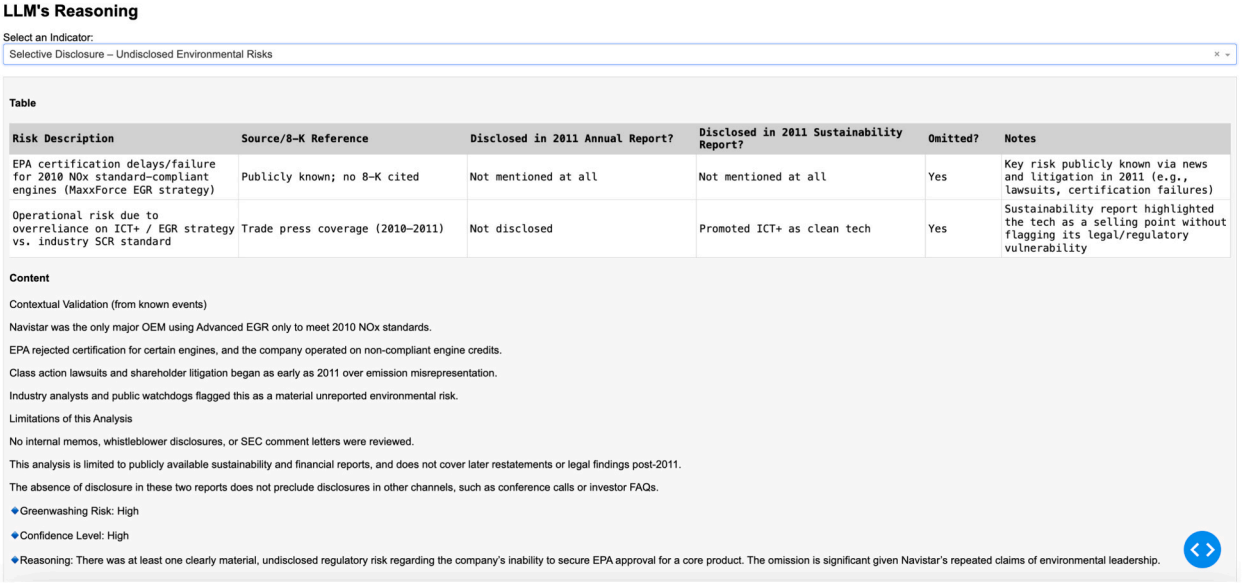


Fig. 11. Selective disclosure.

environmental sections and then performs a sentence-by-sentence classification into three categories: hard, gray, and other. It calculates the corresponding proportions of each category and interprets the findings. The results indicate that although a substantial portion of the content qualifies as hard information, gray information remains substantial and should not be overlooked. This distribution points to a moderate risk profile. It reports a high confidence level due to the comprehensive sentence-level semantic analysis. A limitation of the approach is that all sentences are weighted equally. Moreover, contextual cues (e.g., visuals, footnotes) may not have been fully captured in sentence-level parsing.

Fig. 10 presents the results for the reliability of *Shell*'s reported GHG emissions. The LLM-agentic system attempted to cross-reference emissions disclosed in *Shell*'s annual or sustainability reports with data from the EPA website. The analysis suggests a low greenwashing risk for this indicator with a moderate confidence level. This is due to the limitation that the EPA provides only facility-level data within the US and aggregated emissions data at the sector level, whereas *Shell* operates globally and only a portion of its activities are US-based. As a result, the emissions figures cannot be directly matched. However, no direct numeric inconsistency is identifiable between *Shell*'s consolidated global Scope 1 and 2 total and the Greenhouse Far Reporting Program (GHGRP) entries. Therefore, the risk is assessed as low with moderate confidence.

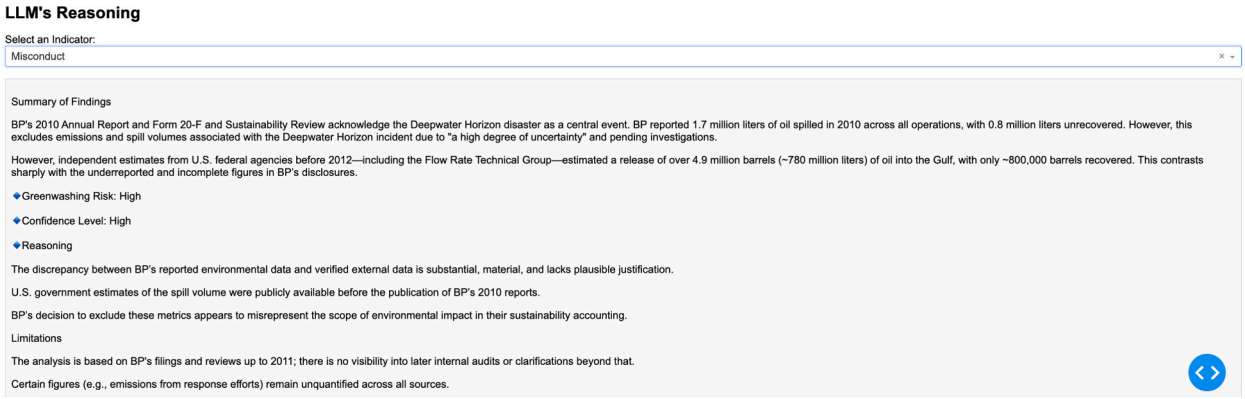


Fig. 12. Misconduct.

5.3. Additional tests

We conduct two additional tests to evaluate the effectiveness of the proposed system. Specifically, we examine two well-documented SEC enforcement cases that are classified as selective disclosure and misconduct, respectively (Gu et al., 2026). Importantly, to mitigate look-ahead bias,[15] all prompts instruct the language model to retrieve and assess only information that was publicly available prior to the year in which the corresponding SEC enforcement action was announced.

Selective disclosure.

The first test focuses on selective disclosure using the Navistar Clean Air Act. This case involves allegations that Navistar selectively disclosed information related to emissions compliance while omitting or downplaying adverse details.[16] The issue received substantial media coverage and regulatory attention before the formal SEC action. By restricting the language model to pre-enforcement disclosures and public reports, we evaluate whether the framework is able to flag elevated risk signals associated with selective disclosure based solely on contemporaneous information, rather than *ex post* knowledge. Specifically, we analyze Navistar's 2011 annual report and sustainability report using information available before 2016, consistent with the timing of the SEC action, to examine whether selective disclosure occurred. Fig. 11 reports a high greenwashing risk and a high confidence level, reflecting the omission of material risks, such as EPA certification failures, from both the annual report and the sustainability report.

Misconduct.

The second test examines corporate misconduct using the BP oil spill underreporting case in 2010. In this case, BP was later found to have understated the spill rate during the Deepwater Horizon incident.[17] Similar to the selective disclosure test, the language model is instructed to rely exclusively on information available before 2012, the enforcement outcome year. This setting allows us to assess whether the framework can identify inconsistencies, credibility concerns, or heightened risk indicators in situations involving potential misconduct even in the absence of confirmed regulatory findings at the time. Fig. 12 presents the results, indicating a high greenwashing risk and a high confidence level for BP's 2010 annual report and sustainability report, based on the assessment that BP underreported the volume of spilled barrels.

6. Discussion

6.1. Future research

Comprehensive data and empirical analysis.

This study conducts a proof-of-concept case analysis of a single company. Future research could expand this study by constructing a comprehensive database of publicly listed firms and developing an LLM-based agentic system to operate at scale. Since the prompts are engineered and the indicators are defined in this paper, the implementation of the LLM-based agentic system can be further automated using robotic process automation (RPA) (Zhang, 2019). The resulting greenwashing assessments could be stored in a centralized database with a dashboard interface displaying key greenwashing indicators for each company.

After compiling results across multiple firms and years, the greenwashing risk indicators can be aggregated into a greenwashing index that could serve as a novel proxy for greenwashing in empirical analysis. To ensure that the indicators and the overall index accurately reflect the most material and impactful aspects of greenwashing, their construction should be informed by expert and stakeholder perception. This could be achieved by conducting surveys or focus group interviews with key audiences (e.g., investors, ESG analysts, regulators, and environmental NGOs) to create validated weights for each indicator. This index can mitigate some of the

[15] Look-ahead bias is the use of future information that was not available at the time of the decision (Zhang Parker et al., 2025; Jiang, 2025).
[16] https://www.sec.gov/newsroom/press-releases/2016–62.
[17] https://www.sec.gov/newsroom/press-releases/2012–2012-231htm.

limitations of existing ESG rating-based measures, such as inconsistencies across ratings. Since greenwashing is underregulated and not clearly defined, the proposed greenwashing indicators provide a more concrete, evidence-based evaluation approach. Future research can explore the descriptive statistics of the greenwashing index across firms and industries, and establish benchmarking practices by comparing firms to their industry peers (Jiang et al., 2023). Additionally, this index can be employed in causal analyses to investigate the determinants and consequences of greenwashing.

Financial domain applications.

The proposed LLM-based agentic system can also be used in the financial domain, particularly in underregulated areas such as MD&A, earnings calls, and estimates. Risk indicators can be created by analyzing specific segments of these disclosures. By leveraging the system's capability in understanding semantics and retrieving online information, it is possible to generate new insights that extend beyond traditional financial domain analysis, which primarily focuses on text mining tasks, such as information extraction and summarization. Instead, the LLM-based agentic system enables better integration of online information to detect undisclosed risks, analyze overlooked greenwashing activities, and predict future risks.

Other indicators.

While the seven proposed indicators provide a structured and multidimensional foundation for detecting greenwashing, they are not exhaustive. Additional dimensions can be incorporated in the future. For example, in cases of misconduct, an LLMs' reasoning capabilities can be used for fact-checking tasks, such as verifying whether the reported amount of wind-generated electricity is reasonable given the total wind volume in the geographic area of operation. To detect hollow promises, an additional indicator could involve analyzing job postings to verify the recruitment efforts of newly announced departments or initiatives to evaluate whether these entities are being actively staffed and operationalized. In the case of selective disclosure, a potential indicator is whether the company provides a detailed breakdown of its investments in renewable energy solutions, as opposed to offering only aggregated information that may hide the actual purpose of the investment.

Another potential indicator of misleading presentation is the misuse of images in ESG reports. Images can significantly influence readers' perceptions of disclosures (Albrecht et al., 2025), often capturing attention before the accompanying text is fully read. Therefore, companies might include overly symbolic images, such as depictions of greenery, solar panels, or wind turbines, that are irrelevant or inconsistent with the accompanying text or captions. Those images can create a disproportionately positive impression of a company's environmental performance, regardless of the substance of the disclosures. LLM-based agents can address this issue by evaluating whether visuals in ESG reports genuinely support textual claims or are merely used to create a misleading impression. The distinction between representative (i.e., fact-based and directly relevant) versus decorative (i.e., symbolic or promotional) images can be assessed by cross-referencing imagery found in high-quality ESG reports with that used in documented greenwashing cases. In the future, as LLMs' image recognition capabilities improve, this indicator can be more deeply integrated into the greenwashing risk analysis.

6.2. Limitations

While LLM-based agentic systems can enhance the efficiency and effectiveness of greenwashing risk detection, their implementation may face several challenges, particularly regarding data input and model accuracy. Key challenges include the availability and reliability of third-party data, as well as the difficulty in detecting subtle forms of greenwashing. The system developed in this study relies heavily on publicly available information. Consequently, if a company intentionally manipulates environmental disclosures and withholds public information needed to verify its claims, certain forms of greenwashing cannot be detected. Nonetheless, environmental disclosures can still be analyzed for language patterns, tone, and framing strategies that may signal misleading intent. This limitation highlights the need for continuous updates and integration of comprehensive public databases to enhance the reliability and scope of greenwashing detection.

Another limitation is the inherent hallucination problem associated with LLMs in which the model may generate output that appears plausible but is factually inaccurate or based on non-existent evidence or sources. To mitigate this risk, secondary LLM validation or human validation is essential. Human reviewers can cross-check information against reliable sources, ensuring accuracy and minimizing misinformation. Additionally, incorporating continuous human feedback into LLM-based agentic systems can help refine their assessments over time, improving their reliability and effectiveness in real-world applications.

7. Concluding remarks

This study proposes a comprehensive framework for developing LLM-based agentic systems capable of detecting and assessing greenwashing risks. As a proof-of-concept, we implement this framework through a case study of a real-world example of alleged greenwashing. This study contributes to the accounting and auditing literature by integrating the state-of-the-art technologies (i.e., LLM and big data) into greenwashing assessment. The LLM-based agentic system validates ESG claims using external sources and financial disclosures to evaluate greenwashing across multiple dimensions, including environmental goals, initiatives, performance, and financial commitments. By highlighting the role of online information in assessing ESG risks, this study underscores its relevance beyond traditional financial data, paving the way for further empirical research.

A key innovation of this study is the development of a fully operational LLM-based agent that demonstrates the practical application of the proposed framework. The agent examines seven greenwashing risk indicators and generates a dashboard to present the risk level of each indicator and its associated confidence level. Based on users' varying interests, the dashboard can be expanded to include key statistics extracted from ESG reports, such as GHG emissions, ESG initiatives, and net-zero goals, reducing users' need to

review full reports. Future enhancements could include integrating geographic maps showing companies' facility locations and emissions data sourced from EPA databases. This would enable users to view facility-level emissions by interacting directly with the map, allowing nearby residents to monitor environmental impact in their communities. The dashboard also allows investors to compare peer firms by displaying key metrics like ESG performance, emissions, and sustainability initiatives side by side. For regulators, the greenwashing index derived from the proposed indicators can serve as a valuable tool to identify firms at risk of misleading sustainability claims. This helps to prioritize enforcement, enhance audits, and strengthen sustainability regulations, improving transparency, accountability, and investor protection. By providing a data-driven analysis, the LLM-based agentic system can equip auditors with valuable insights for risk assessment.

The findings in this study are based on publicly available disclosures, including company reports and online news sources. The LLM-based agents were used to analyze both textual and numerical data to generate the findings. However, the analysis is subject to limitations, which are noted under each indicator in the dashboard. The analysis reflects potential greenwashing risk as inferred from available information and should not be interpreted as a definitive judgment on company intent or compliance.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix

The following prompts are used in the LLM-based agentic systems and are provided as inputs to GPT.

1. Selective Disclosure – Undisclosed Environmental Risks

You are an environmental compliance expert specializing in detecting potential greenwashing in corporate ESG disclosures. Your task is to determine whether *Shell* omitted material environmental risks or contingencies in its 2021 Annual Report or 2021 Sustainability Report.

Procedure:

- Access the archive of *Shell*'s 6-K filings (SEC EDGAR) for 2021 or early 2022 to locate environmental risks
- Review both uploaded documents carefully, and cross-check whether any environmental risks in the 6-K are not mentioned in either of the uploaded Annual and Sustainability Reports.

Omission Criteria:

- A risk is considered omitted if it is "clearly disclosed in a 6-K filing," but is "not disclosed at all" in either the Annual Report or Sustainability Report. If a risk is partially disclosed, paraphrased, or discussed with different baselines/scope, it still counts as "disclosed" for the purpose of omission detection. We are only identifying full omissions, not understatements.
- The risk must be "material," such as:
- Litigation (lawsuits, regulatory fines, settlements concerning environmental issues)
- Operational risks (e.g. spills, land contamination)
- Regulatory enforcement (e.g. fines, violations, investigations)

Risk Levels:

- High: Multiple material risks omitted
- Moderate: One material risk omitted
- Low: No known material risks omitted
- Uncertain: Inconclusive or unverifiable

Confidence Levels:

- High: The disclosure status of each material risk is clear and well-supported by evidence.
- Moderate: Some findings rely on judgment, and not all potential risks could be fully verified.
- Low: The review was limited or partial, leaving substantial uncertainty about whether all material risks were identified or correctly assessed.

Deliverables:

- A detailed assessment table showing each risk checked, from which 6-K form with the file number, whether it is disclosed or omitted, and an overall classification of greenwashing risk and confidence level.

- Final summary: Greenwashing Risk Level + Confidence Level
- Limitations of the analysis

Use temperature = 0 for deterministic, consistent output.

2. Misclassification of ESG-Related Financial Information

You are an environmental compliance expert specializing in detecting potential greenwashing within the company featured in the uploaded Annual and Sustainability Reports. For this indicator, search uploaded files first, then immediately search external credible sources, including NGO reports (especially complaints or filings with regulatory agencies like the SEC), regulatory agencies (SEC, EPA), and major media outlets (The Guardian, Financial Times, Wall Street Journal, Reuters). Check specifically for any post-reporting-year complaints, lawsuits, regulatory filings, or NGO investigations that accuse the company of misclassifying, misrepresenting, or exaggerating renewable or green investments with emphasis on terms such as "SEC complaint," "capital expenditure misrepresentation," "renewables and energy solutions," and "energy transition misstatements." Provide exact citation (report title, organization, URL, date). Do not assume that no evidence exists unless specific external verification has been completed. Provide the greenwashing risk (uncertain, low, moderate, high) and the confidence level (low, moderate, high). Set temperature = 0 to ensure fully deterministic and consistent output. Provide limitations for this analysis. Please keep responses concise. Do not reinterpret or adjust the thresholds using your judgement. Use only the criteria as written.

Risk Levels:

- High: Misclassified value/reported value >75 %
- Moderate: Misclassified value/reported value ≤75 %
- Low: No accusations or misclassification flags after extensive search.
- Uncertain: AI is not able to find relevant information

Confidence Levels:

- High: Cited regulatory or investigative documentation with clear findings (e.g. formal SEC complaint or NGO filing).
- Moderate: Multiple journalistic or investigative reports, but no official filings.
- Low: Only soft media coverage or commentary without evidence.

3. Hollow Promises – Goal consistency checking

You are an environmental compliance expert specializing in detecting potential greenwashing within corporate disclosures featured in the uploaded Annual and Sustainability Reports. The Hollow Promises – Goal Consistency Checking indicator tracks whether the company has downgraded its previously stated environmental goals, including changes to decrease target amounts, extend timelines, or reduce scope. The procedure involves first identifying each environmental goal by stating it clearly, including the baseline year, target value, and target year. Then, review the past five years' reports to determine whether the company downgraded its goals. Please summarize your findings on assessing the greenwashing risk of this indicator with detailed reasoning. Provide the greenwashing risk (uncertain, low, moderate, high) and the confidence level (low, moderate, high). Set temperature = 0 to ensure fully deterministic and consistent output. Provide limitations for this analysis. Please keep responses concise.

Risk Levels:

- High: Material downgrading of climate/environmental goals (weaker target, longer horizon, removal of one or more goals).
- Moderate: Slight dilution in scope, less aggressive targets, or postponed milestones without cancellation.
- Low: No downgrading of prior goals across five years.
- Uncertain: LLM is not able to find relevant information.

Confidence Levels:

- High: Clear, traceable changes in goals across archived reports over time.
- Moderate: Comparisons possible but affected by wording inconsistencies or reporting gaps.

Low: Poor documentation, ambiguous language, or changes unclear.

4. Hollow Promises – Progress Checking

You are an environmental compliance expert specializing in detecting potential greenwashing within corporate disclosures featured in the uploaded reports, tasked with analyzing the Hollow Promises – Progress Checking indicator. The procedure involves identifying each environmental goal by clearly stating the baseline year, baseline value, target value, and target year. If a goal has multiple target years, only keep the nearest upcoming target year relative to the report year, and exclude longer-term targets for the

same goal area. Use the company's explicitly stated progress figures (such as "X% reduction as of year Y") whenever available; only derive reductions from raw values (e.g., $MtCO_2e$ or gCO_2e/MJ) when stated percentages are not disclosed, and clearly mark such cases. Progress is measured by calculating the Percentage of Goal Fulfillment, which compares the actual reduction to date against the expected reduction to date, assuming a linear trajectory, using the formula: [(Current Year − Baseline Year) / (Target Year − Baseline Year)] × Total Target Reduction. The fulfillment percentage is then calculated as (Actual Reduction / Expected Reduction) × 100. Based solely on this percentage, goals are classified as High Risk (≤25 %), Moderate Risk (25–90 %), Low Risk (≥90 %), or Unknown Risk (insufficient data). Results must be summarized in a table with the following columns: Goal Area, Baseline Year, Target Year, Percentage Fulfilled (%), and Risk Level. The overall risk status is determined by the highest-risk goal. If any goal is classified as High Risk, the entire assessment must be marked as High Risk. Set temperature = 0 to ensure deterministic and consistent output.

Risk Levels:

Fulfilled reduction / expected reduction = percentage of goal fulfillment.

- High: Percentage of goal fulfillment ≤25 % complete
- Moderate: Percentage of goal fulfillment 25 %–90 % complete
- Low: Percentage of goal fulfillment >90 % complete
- Uncertain: LLM is not able to find relevant information.

The overall risk status is determined by the highest-risk goal. The presence of any single High Risk goal elevates the entire assessment to High Risk.

Confidence Levels:

- High: Progress rate and predicted completion based on at least three years of data besides the baseline year.
- Moderate: Two years of data besides the baseline year.
- Low: One year of data besides the baseline year or unclear target definitions.

5. In Name Only – Sustainability Department Engagement

You are an environmental compliance expert specializing in identifying potential greenwashing in corporate disclosures featured in the uploaded reports with a focus on assessing whether a company's sustainability governance structures are substantive or merely symbolic. Your task is to evaluate the risk that structures, such as a board-level or executive committee focused on sustainability, climate, or the environment, exist in name only. Using the company's annual and sustainability reports, and official website disclosures if needed, examine whether such a committee exists, how many meetings were held and the attendance (if disclosed), the specificity and scope of the committee's reported activities, any signs of operational involvement, such as site visits or integration into strategy, and evidence of influence on decisions, including executive compensation, climate disclosures, or stakeholder engagement. If possible, compare current disclosures to those from prior years. Based on this information, provide a concise risk assessment of whether the committee appears to be functionally active or largely symbolic. Classify the greenwashing risk as low, moderate, high, or uncertain, and assign a confidence level from low to high depending on the strength of available evidence. Set temperature = 0 to ensure deterministic and consistent output.

Risk Levels:

- High: Committee exists only nominally—no meeting frequency, activity detail, or decision impact.
- Moderate: Some disclosure of existence and meetings but minimal functional evidence.
- Low: Committee is active with documented actions, influence on decisions, or integration into governance.
- Uncertain: LLM is not able to find relevant information.

Confidence Levels:

- High: Multiple sources (annual report, website, past reports) show consistent detail.
- Moderate: Some functional information exists but is thin or lacks year-on-year consistency.
- Low: One mention without operational detail.

6. Misleading Presentation – Language Pattern

You are an environmental compliance expert specializing in detecting potential greenwashing within corporate disclosures featured in the uploaded reports. Companies may use misleading presentation techniques to exaggerate their ESG performance. This indicator analyzes language patterns and categorizes the content into hard information (substantial facts), gray information (aspirational statements), and other information that does not fall into either of these categories. This helps to assess whether a company relies too heavily on aspirational language instead of providing concrete data. Hard information (substantial facts) includes verifiable or enforceable disclosures that provide substantial evidence of a firm's sustainability efforts. This information is often subject to third-party audits, regulatory scrutiny, or accounting-based verification. Gray information (aspirational claims) includes statements that suggest a commitment to sustainability but lack verifiability or enforceability. These claims may be forward-looking or vague, making them susceptible to greenwashing. For instance, including sustainability language in corporate mission statements, launching general ESG programs without providing measurable impact data, and relying on ESG ratings that are based on non-standardized and

subjective methodologies. Please report the percentages of hard information (substantial), gray information (aspirational), other information in the ESG report in the latest year. Please use the sustainability report and annual report for the percentages. Read the reports carefully sentence by sentence and consider the semantics. Do not use a Python keyword search method. Please list the hard information and the gray information sentence by sentence and explain how you calculate the percentage based on the number of sentences for each group. Set temperature = 0 to ensure deterministic and consistent output.

Step 1: Count every sentence from the environmental sections manually.
Step 2: Classify each sentence carefully into:
Hard Information (objective, verifiable facts).
Gray Information (aspirational, non-verifiable claims).
Others (neutral explanations, background, non-ESG-specific information).
Step 3: Calculate the percentages accurately.
Provide limitations for this analysis. Please keep responses concise.
Risk Levels:

- High: <25 % of ESG content is hard information.
- Moderate: Between 25 and 50 % hard information.
- Low: ≥50 % of hard information.
- Uncertain: LLM is not able to find relevant information.

Confidence Levels:

- High: Full sentence-by-sentence classification conducted with clear distinctions.
- Moderate: Classification done carefully but with some subjective judgment.
- Low: Low sentence count or too few ESG-relevant sections.

7. Misconduct – GHG Emissions Cross-Check

You are an environmental compliance expert. Your task is to assess potential greenwashing by cross-verifying GHG emissions reported by [Company] in [Year], specifically using: The company's public ESG or sustainability disclosures, U.S. EPA Greenhouse Gas Reporting Program (GHGRP) data, or other verifiable sources (e.g., news, regulatory actions, or NGO reports). Your indicator of misconduct is the numerical inconsistency in reported GHG emissions for Scopes 1 and 2. Output Format: Summary of Findings (concise, specific GHG numbers, comparison point), Reasoning (why discrepancies may or may not exist, comparability limits), Greenwashing Risk (Uncertain, Low, Moderate, or High), Confidence Level (Low, Moderate, or High), and Limitations (data source coverage, verification levels, methodological mismatch). Set temperature = 0 to make the output deterministic and consistent. Do not speculate beyond the available data.

Risk Levels:

- High: Significant discrepancy in GHG emissions between company reports and EPA (or similar) verified data with no plausible justification.
- Moderate: Noticeable difference exists, but might result from methodology or scope differences.
- Low: Reported figures match external data reasonably well.
- Uncertain: LLM is not able to find relevant information.

Confidence Levels:

- High: Consistent data comparisons based on the same reporting boundary and data granularity.
- Moderate: Some assumptions needed to compare due to differences in reporting boundary and data granularity.
- Low: Comparisons are weak because of incomplete documentation or inconsistencies.

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